sas innovate 2026

Data Driven Programming with CAS in SAS® Viya®

Hands On Workshop

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Agenda

Data Driven Programming with CAS

Practical Examples

Compare to Base SAS

Compare to SAS/Macro

- Introduction to CASL Variables
- CASL Variables, more Detail
- CAS Actions, Briefly
- CAS Action Result Variables
- Load All CSV Files in a Directory to CAS
- Drive Processing from a Table

Introduction to CASL Variables

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Introduction to CASL Variables

SAS Macro vs CASL

CASL Variable

```
%macro conLogic;
  %let aggreg8=N;

  %if &aggreg8=Y %then %do;
    Data public.Agg_Profit ;
    Set public.Orion_Profit ;
      keep Product_Category Customer_Group aggProfit;
      by Product_Category Customer_Group;
      retain aggProfit;
      if first.Customer_Group then AggProfit = 0;
      aggProfit = aggProfit + Profit;
      if last.Customer_Group then output;
    run;


    Proc Print data=public.Agg_Profit;
    run;
  %end;
  %else %do;
    Proc Print data=public.Orion_Profit;
    run;
  %end;
%mend;

%conLogic;
```

```
Proc CAS;
  aggreg8='N';

  if aggreg8='Y' then do;
    datastep.runCode submit / code="
      Data public.Agg_Profit (replace=yes) ;
      Set public.Orion_Profit ;
           keep Product_Category Customer_Group aggProfit;
           by Product_Category Customer_Group;
           retain aggProfit;
           if first.Customer_Group then AggProfit = 0;
           aggProfit = aggProfit + Profit;
           if last.Customer_Group then output;
      run;";

      table.fetch / to=5000
          table={caslib="public" name="aggregatedTable"};
  end;
  else do;
      table.fetch / to=5000
          table={caslib="public" name="detailedTable"};
  end;
quit;
```

CASL Variables, More Detail

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CASL Variables, More Detail

Basic Types

```
Proc CAS;
    /* String */
    custName = "Mary Smith";
    print custName " likes SAS";
    /* Numeric  */
    y = 99;
    print "y plus 50 = " y + 50;
    /* Boolean  */
    exists = true;
    if exists then print exists;
    /* String array */
    colNames = {"make", "model", "type"};
    print "The 2nd column is " colNames[2];
    /* Numeric array */
    percentiles = {5, 25, 50, 75, 95};
    percentiles = percentiles + 99;
    print "The report shows: " percentiles[1:6];
    /* Multi-dimensional array */
    multiDim = { {1, 2}, {3, 4}, {5, 6} };
    print "The 2nd unit of the 3rd pair is " multiDim[3,2];
run;
```

SAS Log

```
Mary Smith likes SAS

y plus 50 = 149

TRUE

The 2nd column is model

The report shows: {5,25,50,75,95,99}

The 2nd unit of the 3rd pair is 6
```

CASL Variables, More Detail

Basic Dictionary Variables

```
proc cas;
    /* Create a dictionary variable */
    myTable = {name="cars", caslib="casuser", replace=true};

    /* Display the structure in the log */
    describe myTable;

    /* Access with an individual key */
    print "Table name is " myTable.name;
quit;
```

SAS Log

```
dictionary ( 3 entries, 3 used);
 [name] string;
 [caslib] string;
 [replace] boolean;

Table name is cars
```

CASL Variables, More Detail

Nested Dictionary Variables

```
proc cas;
    /* Create a dictionary variable */
    myCar = {name="Odyssey", Features = {wheels=4, seats=7, color="red"}};

    /* Display the structure in the log */
    describe myCar;

    /* Access with an individual key */
    print myCar.Features.color;
quit;
```

SAS Log

```
dictionary ( 2 entries, 2 used);
 [name] string;
 [Features] dictionary ( 3 entries, 3 used);
  [wheels] int64_t;
  [seats] int64_t;
  [color] string;


red
```

CAS Actions

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CAS Actions

Briefly

dataStep.**runCode**

simple.**Summary**

regression.**GLM**

table.**loadTable**

table.**tableInfo**

table.**fileInfo**

- Run Inside PROC CAS
- Fine Grain Processing Tools
- Combine with Logic, Variables, and other Actions

CAS Actions

Examples

```
/* Run the FileInfo action */
proc cas;
    table.fileinfo / caslib="public"
quit;
```

Results from table.fileInfo

File Information for root of caslib Public.

Permission	Owner	Group	Name	Size of File in Bytes
-rwxrwxrwx	sas	5001	ERVISITSSYNTHETIC_02_Q2.sashdat	132368
-rwxrwxrwx	sas	5001	simulated_admissions_scoring.sas7bdat	262144
-rwxrwxrwx	sas	5001	ALTITUDEEXPRESS.sashdat	3303792
-rwxrwxrwx	sas	5001	lg_attributes.sas7bdat	131072
-rwxrwxrwx	sas	5001	lookingglass_forecast.sas7bdat	12189696

```
/* Run the Summary action */
proc cas;
    simple.summary
     table={caslib="public" name= "Orion_Profit"
            groupBy="Product_Category"}
     inputs={"Profit"}
     subset={"sum"};
quit;
```

Results from simple.summary

Product_Category=Assorted Sports Articles

Descriptive Statistics for ORION_PROFIT

Column	Sum
Profit	834171.16

Results from simple.summary

Product_Category=Children Sports

Descriptive Statistics for ORION_PROFIT

Column	Sum
Profit	175236.29

Results from simple.summary

Product_Category=Clothes

Descriptive Statistics for ORION_PROFIT

Column	Sum
Profit	722265.005

CAS Action Result Variables

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CAS Action Result Variables

Base SAS vs CASL

CASL Variable

```
*** Write SAS Library Info to a Table ***;

proc datasets lib=sashelp noprint;
   contents data=_all_ out=tabInf;
run;
```

```
*** Write CAS Library Info to a CASL Variable ***;

proc cas;
 table.tableinfo result=tabInf / caslib="public";
run;
```

CAS Action Result Variables

TableInfo Action

```
proc cas;
  table.tableinfo result=tabInf / caslib="public";

    describe tabInf;

quit;
```

SAS Log

```
dictionary ( 1 entries, 1 used);

 [TableInfo] Table ( [19] Rows [25] columns

Column Names:

 [1] Name            [                ] (varchar)
 [2] Label           [                ] (varchar)
 [3] Rows            [                ] (int64)
 [4] Columns         [                ] (int64)
 [5] IndexedColumns  [Indexed Columns ] (int64)
 [6] Encoding        [                ] (char)
 [7] CreateTimeFormatted [Created         ] (varchar)
 [8] ModTimeFormatted [Last Modified   ] (varchar)
 [9] AccessTimeFormatted [Last Accessed   ] (varchar)
 [10] JavaCharSet      [Character Set   ] (char)
 [11] CreateTime       [                ] (double)
 [12] ModTime          [                ] (double)
 [13] AccessTime       [                ] (double)
 [14] Global           [                ] (int32)
 [15] Repeated         [                ] (int32)
 [16] View             [                ] (int32)
 …
```

CAS Action Result Variables

TableInfo Action

```
proc cas;
  table.tableinfo result=tabInf / caslib="public";

    print tabInf.TableInfo;

quit;
```

Table Name	Label	Number of Rows	Number of Columns	Number of Indexed Columns	NLS encoding
LOOKINGGLASS_FORECAST		53999	23	0	utf-8
LG_ATTRIBUTES		918	4	0	utf-8
HOME_EQUITY		5960	18	0	utf-8
VS_BANK		1060038	54	0	utf-8
VS_BANK_SMALL		1060038	17	0	utf-8
ER_VISITS_SYNTHETIC_BIASED_BASIC		10000	22	0	utf-8
ERVISITSSYNTHETIC_01_Q1		500	27	0	utf-8
ERVISITSSYNTHETIC_02_Q2		500	27	0	utf-8
TITANIC3_ID		1309	15	0	utf-8
COMMSDATA		56557	128	0	utf-8
ALTITUDEEXPRESS		10000	23	0	utf-8
PARKS_S		56	6	0	utf-8

CAS Action Result Variables

TableInfo Action

```
proc cas;
  table.tableinfo result=tabInf / caslib="public";

    print tabInf.TableInfo[2,"Name"];

quit;
```

SAS Log

```
LOOKINGGLASS_FORECAST
```

Load all Files in a Directory to CAS

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Load all Files in a Directory to CAS

SAS Macro

```
cas mysess sessopts=(metrics=true);

%macro import_csv_dir(path);

    %let rc = %sysfunc(filename(mydir, &path));
    %let did = %sysfunc(dopen(&mydir));
    %let memcnt = %sysfunc(dnum(&did));

    %do i = 1 %to &memcnt;
        %let name = %qsysfunc(dread(&did, &i));

        %let dsn_name = %qscan(&name, 1, .);

        proc casutil;
            load file="&path/&name" casout="&dsn_name"
            outcaslib="casuser"     importoptions=(filetype="CSV")
            replace ;
        quit;

    %end;

    %let rc = %sysfunc(dclose(&did));

%mend import_csv_dir;

%import_csv_dir(/home/student/Courses/DDPWC);
```

Create a filename to the directory and get the number of files inside.

Iterate over the files and load any CSVs

Load all Files in a Directory to CAS

CASL

Create a caslib to the directory

```
caslib mylib path="/home/student/Courses/DDPWC";


proc cas;
  table.fileinfo result=filesToLoad  / caslib="mylib";

  do i = 1 to dim(filesToLoad.FileInfo);
      fname = filesToLoad.FileInfo[i,'Name'];
      table.loadtable /
                caslib="mylib"
                path=fname;
  end;
quit;
```

Iterate over the files and load any eligible files

Use a Parameter Table to Drive Processing

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Use a Parameter Table to Drive Processing

CASL

Read a Parameter Value from a Table

```
proc cas;
    table.fetch result=OrionParm /
        table={caslib="mylib" name="ParameterTable"
               where="process='Orion_Profit'"};


    whereClause='"' || OrionParm.Fetch[1, 'Condition'] || '"';


    print whereClause;

    simple.summary
        table={caslib="Public" name="Orion_Profit"
               groupBy="Product_Category"
               where=whereClause}
        inputs={"Profit"}
        subset={"sum"};
quit;
```

Use the Parameter in a CAS Action

MYLIB.PARAMETERTABLE

#	Process	Condition
1	Orion_Profit	Product_Line = 'Sports'
2	Nightly_Star	Time > '00:00:00't
3	Orders_Dashb	Date > '31MAR2026'd

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